

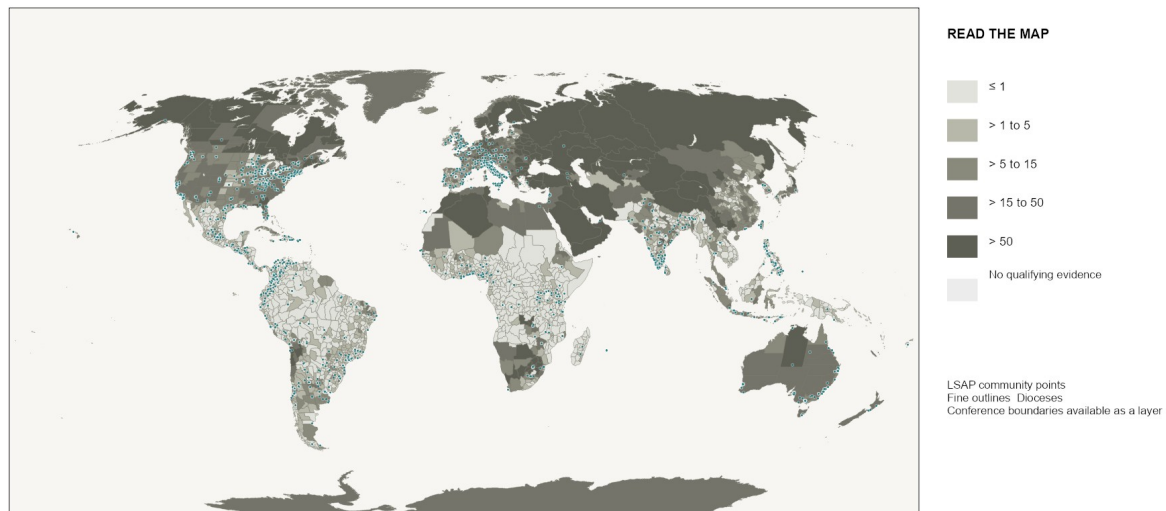
Mapped waste sites across dioceses

This report explains the source data, executed workflow and interpretation of mapped waste sites across dioceses. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

Map P08 | Documentation date 6 October 2026 | Units Distinct sites | Inventory retrieved 6 October 2026

Mapped waste sites across dioceses

Distinct mapped landfill sites assigned to each diocese



Each site is assigned once by greatest footprint overlap. Inventory coverage varies. Zero means no mapped sites, not verified absence.

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LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P08

Figure P08 World overview exported from the completed ArcGIS Pro layout.

Each site is assigned once by greatest footprint overlap. Inventory coverage varies. Zero means no mapped sites, not verified absence.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

1. Query landuse=landfill polygons from seven public regional OpenStreetMap land-use mirrors; retain original JSON, service provenance and the copied OSM_Landfill_Source layer.
2. Deduplicate stable OSM identifiers, exclude missing geometry and repair a working copy. Retain 51,748 unique polygon sites; control status and pollution severity remain unknown.
3. Intersect footprints with copied dioceses using EPSG:6933 equal-area geometry. Assign each distinct site once to the diocese with greatest footprint overlap; resolve ties deterministically.
4. Summarize assigned distinct SiteCount; separately retain IntersectSites for all touching dioceses. Join counts to every diocesan polygon and calculate sites per 1,000 km².
5. Symbolize counts at 1, 5, 15 and 50; retain working footprints for local detail and original source in the geodatabase.

Output fields and files

Field or band	Meaning
SiteCount	Sites uniquely assigned by greatest footprint overlap
IntersectSites	Distinct footprints touching the diocese
Footprint_ha	Unioned clipped footprint area in hectares
AreaPct	Footprint area as percent of full geodesic polygon area
Sites1000km2	Assigned sites per 1000 square kilometres
WasteStory	Partial inventory narrative

Analysis dataset: PollutionAnalysis.gdb/Dioceses_Landfills. Styles/P08.lyrx and Styles/P08.mapx preserve layer and map definitions. Map_Exports/P08_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

Partial polygon inventory; point-only dumps are excluded. Zero means no mapped site. Waste Atlas has not been incorporated: downloadable records and redistribution permission were not verified. Regional source freshness varies, including an older European mirror.

The working inventory contains 51,748 unique sites after removing 2,529 duplicate records. Of these, 50,996 receive a primary diocese assignment and 752 remain unassigned. Raw regional JSON, OSM_Landfill_Source and Landfills_Working remain available. Site control status and pollution severity are unknown. The European mirror is older than most other regions; retrieval date does not imply a uniform data vintage (LSAP Pollution Atlas, 2026, waste_QA.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

References

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